

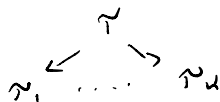
Model selection

Ciaran Evans

Last time: Cross-validation and model selection

Crucial: We must cross-validate the whole model selection process, not just the final chosen model! (model selection is part of fitting process!)

Right way:



Heldout : γ_1

Heldout : γ_2

$\vdots$

Heldout : γ_H

Model selection : $\gamma_2, \dots, \gamma_H \rightarrow \hat{h}_{-1}$

Model selection : $\gamma_1, \gamma_3, \dots, \gamma_H \rightarrow \hat{h}_{-2}$

Model selection : $\gamma_1, \dots, \gamma_{H-1} \rightarrow \hat{h}_{-H}$

the variables
chosen for
each $\hat{h}_{-i}$

may be different!

Today

- ▶ Model selection and hypothesis testing: can I perform valid inference after choosing a model? (activity and discussion)
- ▶ Cross-validation and model selection: activity

Activity 1: Model selection and hypothesis testing

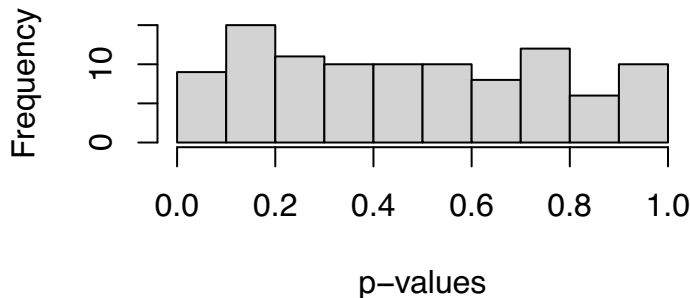
Work on the first class activity (course website)

Activity 1

Under H_0 , $p \sim U(0,1)$

Distribution of reported p-values across all variables:

Histogram of p-values



```
mean(pvals < 0.05)
```

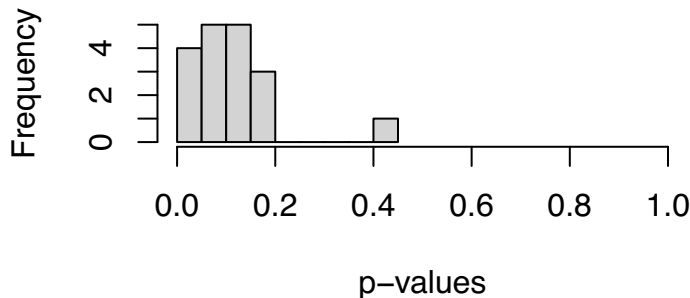
```
## [1] 0.03960396
```

$$P(p_i < \alpha | H_{0i} \text{ is true}) = \alpha$$

Activity 1

Distribution of reported p-values for only the *selected* variables:

Histogram of p-values



```
mean(pvals < 0.05)
```

```
## [1] 0.2222222
```

$$P(p_i < \alpha | H_{0i}, X_i \text{ selected}) > P(p_i < \alpha | H_{0i} \text{ true}) = \alpha$$

Question: Why is testing hypotheses on the selected variables problematic?

The problem

- ▶ Model selection will choose the variables that appear related to the response *in the observed data*
- ▶ If we test only the selected variables, and use the same data for selection and testing, then we have biased the outcome ahead of time: we *know* that these variables must appear related to Y in the data!
- ▶ As a result, we lose type I error control

Data splitting

Let M be some model, and H_0 a hypothesis about M

E.g. $M: \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$

$H_0: \beta_2 = 0$

observe data $\mathcal{T} = \{(x_1, y_1), \dots, (x_n, y_n)\}$

Reject H_0 if $S(\mathcal{T}) > s_0$
(statistic) (rejection threshold)

Type I error control: $P_{M, H_0}(S(\mathcal{T}) > s_0) \leq \alpha$

assuming
 M correctly
specified

assuming H_0 true

model selection & inference:

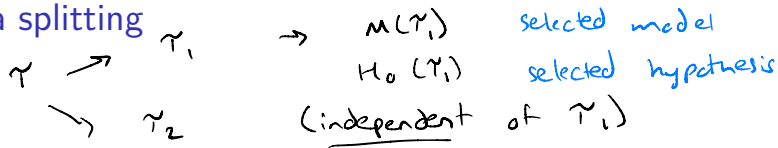
$$M = M(\mathcal{T})$$

$$H_0 = H_0(\mathcal{T})$$

both depend
observed data

$$P_{M(\mathcal{T}), H_0(\mathcal{T})}(S(\mathcal{T}) > s_0 \mid (M(\mathcal{T}), H_0(\mathcal{T})) \text{ selected}) > \alpha$$

Data splitting



Reject $H_0(\gamma_1)$ if $S(\gamma_2) > S_0$

$$P_{M(\gamma_1), H_0(\gamma_1)}(S(\gamma_2) > S_0 \mid (M(\gamma_1), H_0(\gamma_1)) \text{ selected})$$

$$= P_{M(\gamma_1), H_0(\gamma_1)}(S(\gamma_2) > S_0) \leq \alpha \quad \checkmark$$

- Divide data into 2 independent pieces
- Fit model on one piece, test H_0 on other piece

Limitations of automated search methods

- ▶ Not suitable if we have particular inferential questions to test
- ▶ BSS can be computationally expensive
- ▶ Stepwise procedures are greedy and are not guaranteed to find an optimal model (or even a “good” model)
- ▶ Automated procedures **don't** do any of the following:
 - ▶ Check model diagnostics
 - ▶ Experiment with different transformations
 - ▶ Assess model performance
 - ▶ Identify unusual observations
 - ▶ etc.

Activity

Work on the activity (course website).